

## Congruence Criteria — A Decision Guide

Five criteria, two impostors. This page is about choosing the right one; the next page is about the traps.

| Name       | What you must have                                                                     | Picture |
|------------|----------------------------------------------------------------------------------------|---------|
| <b>SSS</b> | All three pairs of SIDES are congruent.                                                |         |
| <b>SAS</b> | Two pairs of sides, and the pair of angles BETWEEN them (the included angle).          |         |
| <b>ASA</b> | Two pairs of angles, and the pair of sides BETWEEN them (the included side).           |         |
| <b>AAS</b> | Two pairs of angles, and a pair of sides NOT between them.                             |         |
| <b>HL</b>  | RIGHT triangles only: the hypotenuses are congruent and one pair of legs is congruent. |         |

### How to Choose — Work in This Order

- 1. Mark the picture first. Put a tick on every pair of sides you know is congruent and an arc on every pair of angles. Do not skip this step — the picture decides the criterion, not the wording.
- 2. Look for the free parts. A SHARED side gives you a congruent pair by the Reflexive Property. VERTICAL angles at a crossing point are congruent. PARALLEL sides give congruent alternate interior angles.
- 3. Count what you have. You need exactly three pairs — no more, no fewer.
- 4. Now name it: three sides → SSS · side-angle-side in that order around the triangle → SAS · angle-side-angle → ASA · angle-angle-side → AAS · right triangle with hypotenuse and a leg → HL.
- 5. Check the word INCLUDED. In SAS the angle sits BETWEEN the two sides. In ASA the side sits BETWEEN the two angles. If it does not, you have AAS — or nothing at all.

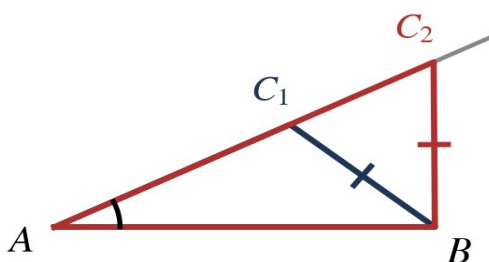
### **The Free Parts Worth Memorizing**

- Shared side → congruent by the Reflexive Property of Congruence.
- Two segments crossing → the vertical angles are congruent.
- A bisected angle → two congruent angles, by the definition of an angle bisector.
- A midpoint or a bisected segment → two congruent segments.
- Parallel sides cut by a transversal → congruent alternate interior angles.
- An isosceles triangle → congruent legs, and congruent base angles.

## **The Two Impostors**

SSA and AAA look like the real criteria. They are not. Here is exactly why.

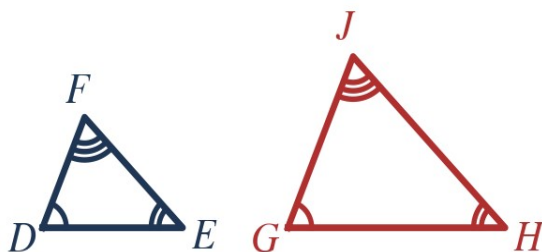
### **SSA — two sides and a NON-included angle**



same two sides and the same non-included angle —  
but two different triangles

Both triangles have the same  $\angle A$ , the same side  $\overline{AB}$ , and the same length for the third side — and they are still different triangles. The second side can swing to either of two positions, so the three parts do not pin the triangle down. That is why SSA proves nothing.

### **AAA — all three angles**



all three angles congruent — but not the same size

Angles control SHAPE, not SIZE.  $\triangle GHJ$  is a scaled copy of  $\triangle DEF$ : every angle matches, yet the triangles are nowhere near congruent. Later you will call this pair SIMILAR — which is a real relationship, just not congruence.

### **The One Exception You Should Know**

- HL also uses two sides and a non-included angle — so why does it work? Because that angle is a RIGHT angle. Once the hypotenuse and one leg are fixed, the Pythagorean Theorem forces the third side, so only one triangle can be built. The swing that ruins SSA is impossible.

Name: \_\_\_\_\_

## Sorting Practice

For each set of information, write SSS, SAS, ASA, AAS, HL — or NONE if the information is not enough. If you write NONE, say why in a few words.

1. Three pairs of congruent sides.
2. Two pairs of congruent sides and the pair of congruent angles between them.
3. Two pairs of congruent angles and the pair of congruent sides between them.
4. Two pairs of congruent angles and a pair of congruent sides not between them.
5. Two pairs of congruent sides and a pair of congruent angles not between them.
6. All three pairs of congruent angles.
7. Two right triangles with congruent hypotenuses and one pair of congruent legs.
8. Two right triangles with one pair of congruent legs and one pair of congruent acute angles.
9. Two triangles share a side; one other pair of sides and the pair of angles between them are congruent.
10. Two segments bisect each other, forming two triangles.
11. A triangle is isosceles and an angle bisector is drawn from the vertex angle.
12. Two pairs of congruent angles only.